MI-22: Resolution

Matthew J. Colbrook
Department of Applied Mathematics and Theoretical Physics
University of Cambridge, Cambridge, United Kingdom
m.colbrook@damtp.cam.ac.uk

11 September 2026

Repository status: Solved. The complete argument received an independent Codex-agent PASS review against the exact catalog target. The original draft was AI-assisted; the reviewed proof below is unchanged. This is independent agent verification, not external human peer review or formal proof certification. Detailed review and scope.

1 MI-22: a certified operator-norm counterexample to the weighted log-majorization

The conjecture is

$$s\big(A^t(A\#_t B)B^{1-t}\big) \prec_{\log} s(AB), \qquad A, B > 0, \quad 0 \le t \le 1, \tag{1}$$

where singular values are decreasingly ordered and $\#_t$ is the weighted geometric mean. It is Conjecture 1.1 of [1]. We disprove even the largest-singular-value inequality. The proof uses explicit rational certificates for two matrix roots; every certificate check is an exact rational comparison.

Theorem 1.1. *A counterexample to* (1) *is*

$$A = \operatorname{diag}\left(256, \frac{1}{256}, 1\right), \qquad B = \begin{pmatrix} 17 & -4 & 0 \\ -4 & 16385 & -8192 \\ 0 & -8192 & 4096 \end{pmatrix}, \qquad t = \frac{1}{8}.$$

In fact,

$$\|A^{1/8}(A\#_{1/8}B)B^{7/8}\|_\infty > 10900, \qquad \|AB\|_\infty < 10200.$$

We first give the elementary error estimate used to turn rational root approximations into a proof.

Lemma 1.2. *If $P, Q \ge mI > 0$ and $0 < \alpha < 1$, then*

$$\|P^\alpha - Q^\alpha\|_\infty \le \alpha m^{\alpha-1}\|P - Q\|_\infty.$$

Proof. The scalar integral representation for x^α, applied by spectral calculus, and the resolvent identity give

$$P^\alpha - Q^\alpha = \frac{\sin(\pi\alpha)}{\pi}\int_0^\infty u^\alpha (uI+Q)^{-1}(P-Q)(uI+P)^{-1}\,du.$$

The integral converges in norm. The norm of the integrand is at most $u^\alpha(u+m)^{-2}\|P-Q\|_\infty$. Differentiating the scalar integral formula, or evaluating this positive integral directly, gives

$$\frac{\sin(\pi\alpha)}{\pi}\int_0^\infty \frac{u^\alpha}{(u+m)^2}\,du = \alpha m^{\alpha-1},$$

proving the estimate. □

Proof of Theorem 1.1. The matrix A is positive definite. The leading principal minors of B are 17, 278529, and 4096, so $B > 0$ as well. Define the rational matrix

$$C = A^{-1/2}BA^{-1/2} = \begin{pmatrix} 17/256 & -4 & 0 \\ -4 & 4194560 & -131072 \\ 0 & -131072 & 4096 \end{pmatrix}.$$

Set $X = C^{1/8}$ and $Y = B^{1/8}$. The left-hand matrix in the theorem is

$$L = A^{5/8} X A^{1/2} Y^7, \qquad A^{5/8} = \operatorname{diag}(32, 1/32, 1), \quad A^{1/2} = \operatorname{diag}(16, 1/16, 1).$$

Here are rational approximations R to X and S to Y, with a common denominator 10^{15}:

$$R = 10^{-15} \begin{pmatrix} 563431071954661 & -4774979464975 & -152620714401404 \\ -4774979464975 & 6722248446399974 & -185449303604146 \\ -152620714401404 & -185449303604146 & 793308473748417 \end{pmatrix},$$

$$S = 10^{-15} \begin{pmatrix} 1417565567728212 & -40084698659651 & -79375309604622 \\ -40084698659651 & 2883518412755210 & -1150490885652445 \\ -79375309604622 & -1150490885652445 & 1157680400969996 \end{pmatrix}.$$

These displayed rational matrices obey the following exact certificate inequalities:

$$B, C > 2^{-10} I, \tag{2}$$

$$\frac{9}{20} I < R < 8I, \qquad \frac{9}{20} I < S < 4I, \tag{3}$$

$$\|R^8 - C\|_F^2 < 10^{-16}, \qquad \|S^8 - B\|_F^2 < 10^{-16}. \tag{4}$$

All positive-definite comparisons in (2)–(3) are verified by positive leading principal minors. The two comparisons in (4) are sums of nine rational squares. The accompanying certificate verifier performs precisely these operations, without floating-point arithmetic, and its full input is printed above.

Since $(9/20)^8 > 2^{-10}$, both R^8 and S^8 exceed $2^{-10} I$. Apply Lemma 1.2 with $\alpha = 1/8$ and $m = 2^{-10}$. Because $\alpha m^{\alpha - 1} = 2^{23/4} < 64$, equations (2)–(4) imply

$$\|X - R\|_\infty < \varepsilon, \qquad \|Y - S\|_\infty < \varepsilon, \qquad \varepsilon = 64 \cdot 10^{-8}.$$

Also $\|X\|_\infty < 8$ and $\|Y\|_\infty < 4$: for example, the exact traces of C and B are respectively smaller than 8^8 and 4^8. Together with (3), the telescoping identity for $Y^7 - S^7$ gives

$$\|Y^7 - S^7\|_\infty \le 7 \cdot 4^6 \varepsilon.$$

For the rational matrix $\widetilde{L} = A^{5/8} R A^{1/2} S^7$, it follows that

$$\begin{aligned} \|L - \widetilde{L}\|_\infty &\le 32 \cdot 16\, \varepsilon \bigl(4^7 + 8 \cdot 7 \cdot 4^6\bigr) \\ &= \frac{6291456}{78125} < 100. \end{aligned}$$

One further exact rational multiplication yields

$$\widetilde{L}_{12} > 11000. \tag{5}$$

Consequently $\|L\|_\infty \ge |L_{12}| > 11000 - 100 = 10900$.

On the other hand,

$$AB = \begin{pmatrix} 4352 & -1024 & 0 \\ -1/64 & 16385/256 & -32 \\ 0 & -8192 & 4096 \end{pmatrix},$$

and its exact squared Frobenius norm is

$$\|AB\|_F^2 = \frac{6807858741265}{65536} < 10200^2.$$

Thus $\|AB\|_\infty \le \|AB\|_F < 10200 < 10900 < \|L\|_\infty$, contradicting the first inequality required by log-majorization. $\square$

Remark 1.3 (Reproducibility and the role of computation). High-precision calculations were used to propose the rational matrices R, S, but no such calculation is trusted by the proof. The finite certificate consists of (2)–(4), (5), and the Frobenius-norm comparison; all are verified with integer fractions. The error estimate bridges these exact finite checks to the true matrix roots. This is a rigorous computer-assisted counterexample, not a claim based only on a numerical singular-value comparison.

References

[1] Mohammad M. Ghabries, Hassane Abbas, Bassam Mourad, and Abdallah Assi. New log-majorization results concerning eigenvalues and singular values and a complement of a norm inequality. arXiv:2105.13356, 2021. Conjectures 1.1 and 2.1. URL: https://arxiv.org/abs/2105.13356.